Recoil and explosions

antocciphysics | TEACHER KEY | CPA

Learning target: Use momentum conservation for an isolated system.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 2; 3 course days later.*
15 kg·m/s left.

02. Explain a worked example (worked)
Initial total zero; after arrows +4 and -4 kg·m/s cancel.

03. Look and explain (model)
Equal magnitudes, opposite directions; total stays zero.

04. Build the solution (practice)
Momentum 4 kg·m/s right needed; speed 2 m/s.

05. Use the idea (reason)
Stored internal energy can become kinetic energy; momentum conservation does not require kinetic energy conservation.

06. Choose a method (mixed) *Review: Unit 2, Day 9; 12 course days later.*
Equal fall times; faster ball travels 6 times farther horizontally.

07. Exit check (exit)
6 kg·m/s right.

08. STRETCH: Apply it another way (stretch)
Second mass 3 kg; equal opposite momenta +6 and -6 kg·m/s.

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01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 2; 3 course days later.*
v=-5 m/s; after reversal p=+15 kg·m/s.

02. Explain a worked example (worked)
Initial total zero; after arrows +4 and -4 kg·m/s cancel.

03. Represent and interpret (model)
Equal magnitudes, opposite directions; total stays zero.

04. Solve and justify (practice)
60 v+24=0; v=-.40 m/s.

05. Analyze the reasoning (reason)
Stored internal energy can become kinetic energy; momentum conservation does not require kinetic energy conservation.

06. Choose a method (mixed) *Review: Unit 2, Day 9; 12 course days later.*
Speed 20 m/s. Need an initial position plus time and motion history/conditions; velocity alone is insufficient.

07. Exit check (exit)
v=-6 m/s; KE 81+54=135 J from released internal energy.

08. STRETCH: Apply it another way (stretch)
Person -0.5 m/s; relative board speed 3.5 m/s. Velocities in conservation must use one frame.